Chengsong Huang

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Education

Washington University in St. Louis Sep. 2023 - Present
Advisor: Prof. Jiaxin Huang St. Louis, MO
Ph.D. in Computer Science (GPA 4.0/4.0)

Fudan University Sep. 2019 - Jun. 2023
Software School of Fudan University Shanghai, China
B.S. in Software Engineering (GPA 3.82/4.0) rank 1%

Research Interests

My research focuses on advancing the capabilities and accessibility of Large Language Models (LLMs). My interests include:

- **Data-Driven Self-Evolution:** Designing frameworks that enable LLMs to evolve themselves through the data and feedback they generate. This includes self-evolving training loops that synthesize their own training data and curricula from zero human-labeled data, self-rewarding models that turn their own reasoning into supervision, and environments that co-evolve with the agents trained in them. Representative works: R-Zero (ICLR 2026), a reasoning LLM that self-evolves from zero data, and EnvHarness (Google Research), which awakens static environments for agent learning.
- **Efficient LLMs:** Developing and analyzing methods to make LLMs more computationally efficient. My experience includes work on parameter-efficient fine-tuning (PEFT), speculative decoding, and adaptive test-time scaling to accelerate training and inference.

(Co-)First-Author Preprints

** stands for equal contribution or alphabetical ordering.*

Data-Driven Self-Evolution

- **Huang, C.**, Wang, Z., Han, R., Yan, J., Chen, Y., CuiZhu, Z., Jiang, K., Xia, P., Yu, H., Zhuang, Y., Ming, Y., Pan, J., Dalvi Mishra, B., Huang, J., Gokturk, B., Pfister, T. and Lee, C.-Y. "EnvHarness: Awakening Static Worlds for Agent Learning". arXiv preprint. 2026.

Others

- Qian, Q.*, **Huang, C.***, Xu, J., Lv, C., Wu, M., Liu, W., Wang, X., Wang, Z., Huang, Z., Tian, M., Xu, J., Hu, K., Wang, H.-D., Hu, Y., Huang, X. and Zheng, X. "Benchmark^2: Systematic Evaluation of LLM Benchmarks". arXiv preprint. 2026.
- Gu, C.*, **Huang, C.***, Zheng, X., Chang, K.-W. and Hsieh, C.-J. "Watermarking Pre-trained Language Models with Backdooring". arXiv preprint. 2022.

(Co-)First-Author Published Papers

For a complete list of publications, please visit my Google Scholar profile.

** stands for equal contribution or alphabetical ordering.*

Data-Driven Self-Evolution

- **Huang, C.**, Yu, W., Wang, X., Zhang, H., Li, Z., Li, R., Huang, J., Mi, H. and Yu, D. "R-Zero: Self-Evolving Reasoning LLM from Zero Data". ICLR 2026.
- **Huang, C.**, Liu, H., Zheng, T., Dai, R., Huang, L., Li, J., Li, Z., Wei, Z., Meng, Y. and Huang, J. "G-Zero: Self-Play for Open-Ended Generation from Zero Data". NeurIPS 2026 **(Spotlight)**. **Best Paper Award**, NeurIPS 2026 Workshop on Towards Test-Time Continual Learning Agents (TTCL).
- He, Y.*, **Huang, C.***, Li, Z.*, Huang, J. and Yang, Y. "VisPlay: Self-Evolving Vision-Language Models from Images". CVPR 2026.

Efficient LLMs

- Zheng, T.*, **Huang, C.***, Dai, R.*, He, Y., Liu, R., Ni, X., Bao, H., Wang, K., Zhu, H., Huang, J., Huang, F. and Huang, H. "Parallel-Probe: Towards Efficient Parallel Thinking via 2D Probing". ICML 2026.
- **Huang, C.**, Huang, L., Leng, J., Liu, J. and Huang, J. "Efficient Test-Time Scaling via Self-Calibration". ICLR 2026.
- **Huang, C.**, Zheng, T., Huang, L., Li, J., Liu, H. and Huang, J. "RelayLLM: Efficient Reasoning via Collaborative Decoding". Findings of EMNLP 2026.
- **Huang, C.**, Huang, L. and Huang, J. "Divide, Reweight, and Conquer: A Logit Arithmetic Approach for In-Context Learning". EACL 2026.
- **Huang, C.***, Liu, Q.*, Lin, B. Y.*, Pang, T., Du, C. and Lin, M. "LoraHub: Efficient Cross-Task Generalization via Dynamic LoRA Composition". COLM 2024.

Others

- Lin, B. Y.*, **Huang, C.***, Liu, Q., Gu, W., Sommerer, S. and Ren, X. "On Grounded Planning for Embodied Tasks with Language Models". AAAI 2023.

Experience

Google Cloud AI Feb. 2026 - present
Student Researcher (Mentor: Zifeng Wang)
EnvHarness: Awakening Static Worlds for Agent Learning

Tencent AI Lab Seattle Jun. 2025 - Aug. 2025
Research intern (Mentor: Wenhao Yu)
R-Zero: Self-Evolving Reasoning LLM from Zero Data

Sea AI lab (SAIL) Jan. 2023 - Jun. 2023
Research intern (Mentor: Qian Liu)
LoraHub: Efficient Cross-Task Generalization via Dynamic LoRA Composition

USC INK lab May. 2022 - Jan. 2023
Research intern (Mentor: Xiang Ren, Bill Yuchen Lin)
On Grounded Planning for Embodied Tasks with Language Models

Open-Source Impact

- Author and maintainer of open-source research code with **2,200+ GitHub stars** in total, including R-Zero (850+ stars), LoraHub (650+ stars), EnvHarness (600+ stars), VisPlay, RelayLLM, G-Zero and Self-Calibration.

Service

Conference Reviewer: ARR 2024-present, ICLR 2025-present, COLM 2024-present, AAAI 2026, CVPR 2026, ICML 2026.